



StreamStats Version 3.0

Flow Statistics Ungaged Site Report

Date: Mon Dec 21, 2015 11:05:26 AM GMT-5

Study Area: Massachusetts

NAD 1983 Latitude: 42.2084 (42 12 30)

NAD 1983 Longitude: -71.24 (-71 14 24)

Drainage Area: 1.36 mi²

Low Flows Basin Characteristics			
100% Statewide Low Flow WRIR00 4135 (1.36 mi ²)			
Parameter	Value	Regression Equation Valid Range	
		Min	Max
Drainage Area (square miles)	1.36 (below min value 1.61)	1.61	149
Mean Basin Slope from 250K DEM (percent)	2.259	0.32	24.6
Stratified Drift per Stream Length (square mile per mile)	0.059	0	1.29
Massachusetts Region (dimensionless)	0	0	1

Warning: Some parameters are outside the suggested range. Estimates will be extrapolations with unknown errors.

Probability of Perennial Flow Basin Characteristics			
100% Perennial Flow Probability (1.36 mi ²)			
Parameter	Value	Regression Equation Valid Range	
		Min	Max
Drainage Area (square miles)	1.36	0.01	1.99
Percent Underlain By Sand And Gravel (percent)	10.40	0	100
Percent Forest (percent)	66.65	0	100
Massachusetts Region (dimensionless)	0	0	1

Bankfull Flows Basin Characteristics			
100% Bankfull Statewide SIR2013 5155 (1.36 mi ²)			
Parameter	Value	Regression Equation Valid Range	
		Min	Max
Drainage Area (square miles)	1.36	0.6	329
Mean Basin Slope from 10m DEM (percent)	6.671	2.2	23.9

Low Flows Statistics						
Statistic	Value	Unit	Prediction Error (percent)	Equivalent years of record	90-Percent Prediction Interval	
					Min	Max
D50	1.31	ft ³ /s				
D60	0.84	ft ³ /s				
D70	0.44	ft ³ /s				
D75	0.32	ft ³ /s				

D80	0.23	ft3/s				
D85	0.16	ft3/s				
D90	0.1	ft3/s				
D95	0.0558	ft3/s				
D98	0.0346	ft3/s				
D99	0.0241	ft3/s				
M7D2Y	0.0631	ft3/s				
AUGD50	0.17	ft3/s				
M7D10Y	0.0199	ft3/s				

<http://pubs.usgs.gov/wri/wri004135/> (<http://pubs.usgs.gov/wri/wri004135/>)

Ries_ K.G._ III_ 2000_ Methods for estimating low-flow statistics for Massachusetts streams: U.S. Geological Survey Water Resources Investigations Report 00-4135_ 81 p.

Probability of Perennial Flow Statistics						
Statistic	Value	Unit	Prediction Error (percent)	Equivalent years of record	90-Percent Prediction Interval	
					Min	Max
PROBPEREN	0.84	dim	0.3		0.57	0.8

http://pubs.usgs.gov/sir/2006/5031/pdfs/SIR_2006-5031rev.pdf (http://pubs.usgs.gov/sir/2006/5031/pdfs/SIR_2006-5031rev.pdf)
Bent_ G.C._ and Steeves_ P.A._ 2006_ A revised logistic regression equation and an automated procedure for mapping the probability of a stream flowing perennially in Massachusetts: U.S. Geological Survey Scientific Investigations Report 2006-5031_ 107 p.

Bankfull Flows Statistics						
Statistic	Value	Unit	Prediction Error (percent)	Equivalent years of record	90-Percent Prediction Interval	
					Min	Max
BFWDTH	16.8	ft	21			
BFDPTH	1.03	ft	20			
BFAREA	17.1	ft2	29			
BFFLOW	44.5	ft3/s	55			

<http://pubs.usgs.gov/sir/2013/5155/> (<http://pubs.usgs.gov/sir/2013/5155/>)

Bent_ G.C._ and Waite_ A.M._ 2013_ Equations for estimating bankfull channel geometry and discharge for streams in Massachusetts: U.S. Geological Survey Scientific Investigations Report 2013-5155_ 62 p._

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URL: http://streamstatsags.cr.usgs.gov/v3_beta/FTreport.htm

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StreamStats Version 3.0
Basin Characteristics Ungaged Site Report

Date: Mon Dec 21, 2015 9:55:01 AM GMT-5

Study Area: Massachusetts

NAD 1983 Latitude: 42.2084 (42 12 30)

NAD 1983 Longitude: -71.24 (-71 14 24)

Label	Value	Units	Definition
DRNAREA	1.36	square miles	Area that drains to a point on a stream
STRMTOT	2.48	miles	Total length of mapped streams in basin
DRFTPERSTR	0.059	square mile per mile	Area of stratified drift per unit of stream length
MAREGION	0	dimensionless	Region of Massachusetts 0 for Eastern 1 for Western
FOREST	66.65	percent	Percentage of area covered by forest
CRSDFT	10.4	percent	Percentage of area of coarse-grained stratified drift
BSLDEM10M	6.671	percent	Mean basin slope computed from 10 m DEM
BSLDEM250	2.259	percent	Mean basin slope computed from 1:250K DEM
ACRSDF	0.15	square miles	Area underlain by stratified drift
LC11IMP	3.26	percent	Percentage of impervious area determined from NLCD 2011 impervious dataset
LC11DEV	16.3	percent	Percentage of land-use from NLCD 2011 classes 21-24
ELEV	277	feet	Mean Basin Elevation
PRECPRIS00	47.8	inches	Basin average mean annual precipitation for 1971 to 2000 from PRISM
LAKEAREA	0.26	percent	Percentage of Lakes and Ponds
OUTLETX	221465	State plane coordinates	Basin outlet horizontal (x) location in state plane coordinates
OUTLETY	884245	State plane coordinates	Basin outlet vertical (y) location in state plane coordinates
MAXTEMP	15.1	degrees	Mean annual maximum air temperature over basin area, in degrees Centigrade
WETLAND	14.09	percent	Percentage of Wetlands
CENTROIDX	220751.6	State plane coordinates	Basin centroid horizontal (x) location in state plane coordinates
CENTROIDY	885474.1	State plane coordinates	Basin centroid vertical (y) location in state plane units

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URL: http://streamstats.cr.usgs.gov/v3_beta/BCreport.htm

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